Adversarial Robustness of CNNs:
Attacks, Explainability, and Defense on Caltech 101

MIA Robotics — AI Team Training '27 — Task 5, Phase II

Investigating Professor Moriarty's "Digital Cloak"

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1. Introduction

This report documents a full pipeline for studying adversarial vulnerability in convolutional neural networks, following the five-phase structure of the assignment: (1) transfer-learning model preparation, (2) adversarial attack implementation, (3) explainability (XAI) tooling, (4) forensic analysis of individual failures, and (5) a defensive countermeasure with quantitative evaluation. All model code was implemented in PyTorch, using a ResNet-34 backbone fine-tuned on the Caltech 101 dataset.

2. Phase 1 — Model Preparation and Fine-Tuning

A ResNet-34 model pre-trained on ImageNet was adapted to Caltech 101 (102 classes) using transfer learning. All convolutional layers were frozen except the final block (layer4), and the fully-connected classification head was replaced with a new linear layer matching the number of target classes. Only the unfrozen parameters (layer4 and the new head) were optimized using Adam (lr = 0.001) with cross-entropy loss.

After 10 training epochs, the model reached a peak validation accuracy of **94.12%**, stabilizing in the 91-94% range — a strong baseline for the subsequent attack and defense experiments.

Epoch	Loss	Val. Accuracy
1	1.0848	85.08%
3	0.1652	90.21%
5	0.0840	91.82%
7	0.0438	94.12%
10	0.0834	93.26%

3. Phase 2 — Adversarial Attack: FGSM

The Fast Gradient Sign Method (FGSM) was implemented to generate adversarial examples. FGSM perturbs an input image in the direction that maximizes the model's loss, using only the sign of the gradient so that every pixel is perturbed by exactly the same magnitude, epsilon:

```
x_adv = x + epsilon * sign(∇ₓ Loss(f(x), y))
```

A key implementation detail was correctly clamping the perturbed image to the valid range in normalized pixel space (approximately [-2.12, 2.25] given the ImageNet normalization statistics), rather than [0, 1], since the latter silently destroyed the image regardless of the attack strength.

Effect of epsilon on validation accuracy (full validation set):

Epsilon	0	0.001	0.005	0.01	0.03	0.05	0.1	0.2
Accuracy	93.26%	66.88%	47.93%	48.39%	53.05%	46.03%	10.43%	0.86%

Clean accuracy (93.26%) confirms the epsilon = 0 control case is correctly implemented. Accuracy drops sharply even at very small perturbations (epsilon = 0.001), and the model is almost fully defeated at epsilon = 0.1-0.2. The minor non-monotonic fluctuation around epsilon = 0.03 is an expected property of single-step gradient attacks on non-linear decision boundaries: a larger step in the same fixed gradient direction can occasionally overshoot into a different (sometimes correct) decision region, rather than monotonically increasing error. Based on this sweep, the **minimum epsilon producing consistent misclassification is approximately 0.005**, and **epsilon = 0.1** was selected as the strongest attack for later evaluation.

4. Phase 3 — Explainability (XAI)

Two complementary explainability techniques were implemented to visualize model decisions:

Saliency Maps (Vanilla Gradients): the gradient of the predicted class score with respect to each input pixel, taking the absolute value and maximizing across RGB channels. This operates at the pixel level and is therefore sensitive to noise, producing scattered, speckled maps.

Grad-CAM: computed from the activations and gradients of the last convolutional block (layer4) rather than raw pixels, producing smooth, semantically meaningful heatmaps. Two implementation issues were identified and corrected during development: (1) the input tensor must retain `requires_grad=True` for backward hooks to fire correctly, and (2) using the raw class logit (rather than the cross-entropy loss) as the backward target avoids vanishing gradients when the model is highly confident.

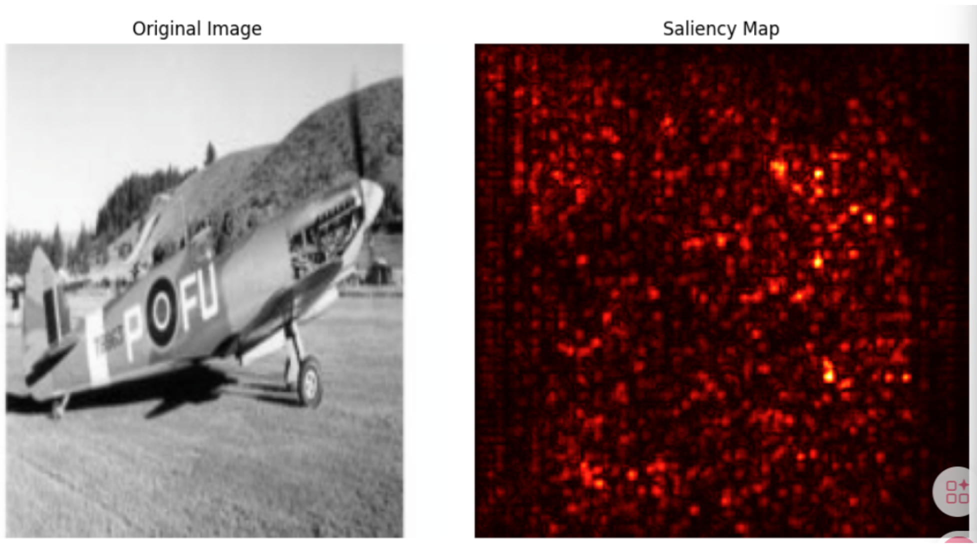

Figure 1. Saliency map (vanilla gradients) on a clean, correctly-classified airplane image. Note the scattered, high-frequency noise pattern.

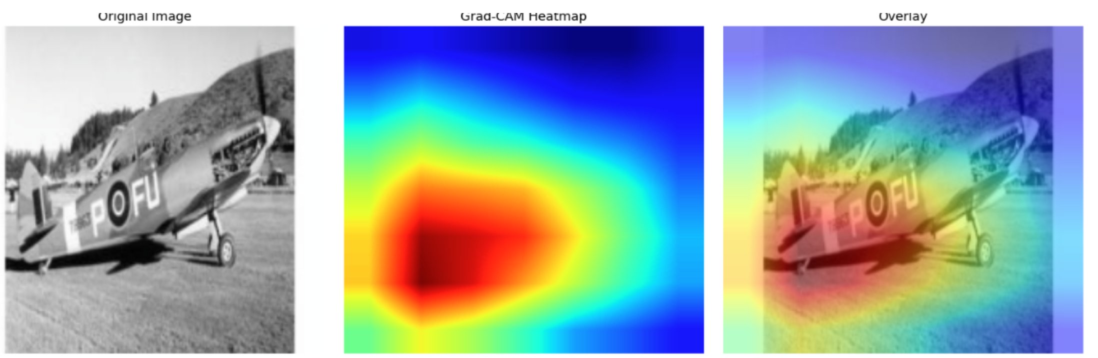

Figure 2. Grad-CAM on the same image, cleanly localizing the airplane body/nose — a much smoother and more interpretable region than the saliency map.

5. Phase 4 — Forensic Analysis

For each example, a full forensic record was assembled combining: the clean prediction and its XAI maps, the attacked image with the resulting misclassification and a visualization of the adversarial noise itself, and — critically — the XAI maps computed with respect to the model's *new incorrect prediction* (rather than the true label), in order to diagnose why the model produced that specific wrong answer.

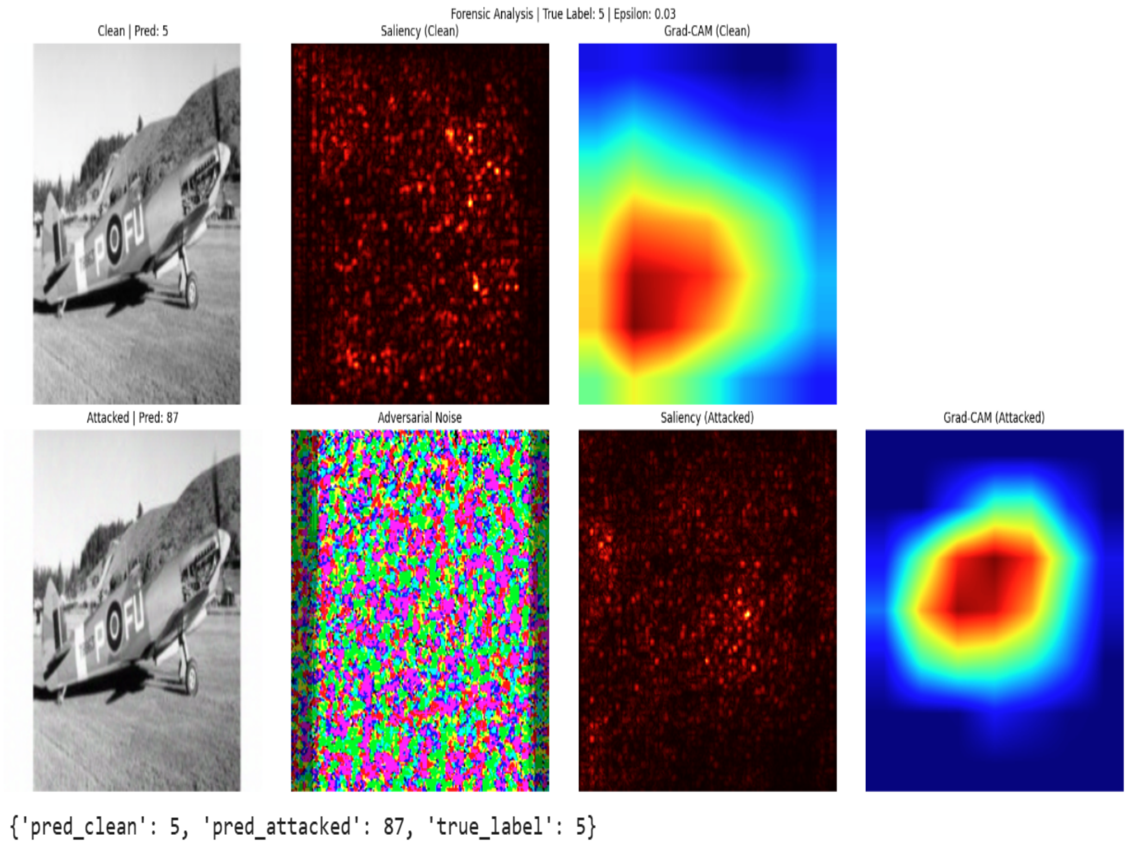

Figure 3. Forensic analysis at epsilon = 0.03. Top row: clean image, saliency map, Grad-CAM (correct prediction, class 5). Bottom row: attacked image (visually identical to the human eye), the adversarial noise pattern, and saliency/Grad-CAM for the new incorrect prediction (class 87).

This experiment, and two further examples from different classes (76 and 81), confirm that the attack generalizes across categories: all three correctly-classified, high-confidence clean images were successfully flipped to incorrect predictions at epsilon = 0.03.

Failure mechanism: the misclassification is not caused by a large volume of noise, but by a small, precisely-directed perturbation computed from the model's own gradient. The Grad-CAM comparison shows that this perturbation is enough to shift the model's spatial attention to a different, smaller region of the image, which is sufficient to flip the decision entirely — despite the perturbed image being visually indistinguishable from the original to a human observer. This is precisely the "Digital Cloak" phenomenon described in the assignment brief.

6. Phase 5 — Defense: Defensive Distillation

Defensive Distillation was implemented as the hardening technique. A Teacher model (the fine-tuned ResNet-34 from Phase 1) produces "soft labels" using a softened softmax with temperature T: $\mathrm{softmax}_T(z_i) = \exp(z_i/T) / \sum_j \exp(z_j/T)$. A Student model with an identical architecture is then trained from scratch to match these soft probability distributions using KL-divergence loss, rather than the original hard labels. Both Teacher and Student logits were divided by $T = 4$ before the softmax/log-softmax step to compare distributions on the same scale, and the resulting loss was multiplied by T-squared to counteract the corresponding shrinkage of the gradient magnitude introduced by the temperature scaling.

Distillation loss decreased steadily over 10 training epochs, from 9.98 to 0.79, and the Student reached 90.67% clean validation accuracy — close to the Teacher's baseline performance.

6.1 The Showdown: Quantitative Evaluation

Model	Clean Accuracy	Adversarial Accuracy (FGSM, eps=0.1)	Accuracy Drop
Teacher (Phase 1)	87.96%	15.90%	72.06%
Student (Distilled)	90.67%	11.75%	78.92%

The distilled Student model shows a slightly higher clean accuracy than the Teacher (90.67% vs. 87.96%), but a **larger** accuracy drop under the same FGSM attack (78.92% vs. 72.06%). In other words, Defensive Distillation did **not** improve robustness in this experiment — it made the model marginally more vulnerable to the attack.

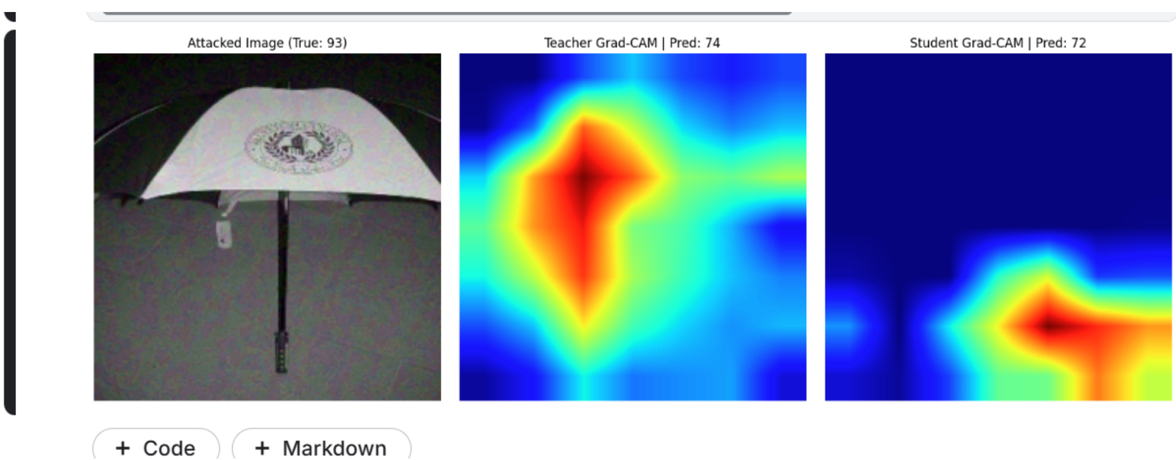

Figure 4. Grad-CAM comparison under the same FGSM attack (eps=0.1) on a held-out image (true class: umbrella, 93). Teacher predicts class 74, focusing on the umbrella's central pole/handle. Student predicts a different wrong class (72), attending to an unrelated region of the image entirely.

6.2 Interpretation

This result is consistent with the well-documented critique of Defensive Distillation by Carlini and Wagner (2016), who demonstrated that the technique does not increase genuine model robustness. Instead, the very small gradients induced by training at high temperature cause **gradient masking**: the gradient signal that a gradient-based attack (such as FGSM) relies on becomes numerically small and uninformative, giving the false impression of robustness, while the model's underlying decision boundaries remain just as exploitable — and, as observed here, can even be exploited slightly more easily under a fixed-strength attack. The Grad-CAM comparison in Figure 4 supports this: under attack, the Student model attends to a region of the image that is *not* semantically related to the true or predicted class, suggesting that it has not learned genuinely more robust features, only a smoother loss surface.

7. Conclusion

- A ResNet-34 model fine-tuned on Caltech 101 reached ~93% validation accuracy, but could be reliably fooled by FGSM perturbations as small as epsilon = 0.005, imperceptible to the human eye.
- Grad-CAM proved substantially more interpretable than vanilla saliency maps for diagnosing failures, revealing that adversarial perturbations shift the model's spatial attention to different, often semantically irrelevant, regions of the image.
- Defensive Distillation, despite a research-motivated implementation with correct temperature scaling, did not produce genuine robustness against FGSM and is best explained as a case of gradient masking, consistent with prior literature (Carlini & Wagner, 2016).
- Future work could evaluate Adversarial Training or input-transformation defenses, which are not susceptible to gradient masking in the same way, as alternative countermeasures.